DL Project Proposal

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1 Title

Mechanistic Interpretability of Reasoning Models with Crosscoder Model Diffing

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3 Project Summary

Reasoning models like OpenAI's o1 excel at a number of tasks and are the current SOTA on verifiable domains like math and coding [1, 2]. These models are able to "think" for longer during inference time and often have chains of thought where they break down problems into steps and sometimes backtrack when they don't have a correct answer [3].

Yet, little is understood about what changes in the models are needed for them to be able to perform as well as they do. In particular, how they "decide" to backtrack or what they have found they "believe" to be the correct answer is a research area that is quickly evolving and has little work published on it [4, 5].

Mechanistic interpretability is a subfield of Deep Learning focused on understanding how models work. The lectures on visualization covered saliency maps for vision models, which are results of mechanistic interpretability research. Some of the landmark discoveries on LLMs over the last few years have focused on extracting interpretable features from transformer models [6, 7, 8]. Much of the work examines superposition, the idea that individual neurons can encode information for multiple different unrelated concepts. The representational space of models is much larger than the number of neurons, and this space is quite sparse.

Sparse autoencoders (SAEs) are special networks trained to disentangle overlapping features into a much sparser and more interpretable space. They are trained on the activations of other models when varied output is passed through them [6]. SAEs focus on a single block in a transformer. Crosscoders are a newer technique designed to interpret a model's latent representations across multiple layers of a network. They have also been applied to diff what models represent at different points in training or even between separate models [9, 10].

4 Goal and Approach

Research goal: Start to understand what changes in a model when it is fine-tuned to reason.
Approach Overview: Train a crosscoder on the base model and the fine-tuned model and perform analogous interpretability experiments on both.

A recent paper, "s1: Simple test-time scaling," [11] by Niklas out of Li Fei-Fei's group at Stanford, sought to find "the simplest approach to achieve test-time scaling and strong reasoning performance." They created a dataset, s1k, of 1k samples of reasoning traces of DeepSeek R1 in response to a broad set of questions. They then fine-tuned Qwen2.5-32B-Instruct in only 26 minutes on a cluster of 16 H100s into a model they call s1-32B, which was able to exceed o1-preview on competition math questions. The base model, fine-tuned model, dataset, and code are all publicly available.

The research direction we have been exploring is applying the s1k dataset to fine-tune an even smaller model (somewhere in the ballpark of 350M to 7B parameters) and evaluating the difference in performance on reasoning tasks before and after (AIME2024, Math-500, GPQA Diamond).

We then plan to train crosscoders on both the base model and the SFT model to diff the models and analyze what has changed [12].

5 Dataset

https://huggingface.co/datasets/simplescaling/s1K-1.1

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